

HAT-P-32b

R-0245

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_240920 · created 2026-07-17T05:41:45+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460573.774 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1721 +/- 0.0094
Transit depth (Rp/Rs)^2	2.96 +/- 0.32 [%]
Orbital Inclination (inc)	84.0 +/- 1.7
Airmass coefficient 1 (a1)	1.115 +/- 0.01
Airmass coefficient 2 (a2)	-0.103 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	None
Optimal Aperture	3.63
Optimal Annulus	9.5
Transit Duration (day)	0.1025 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1721 $\pm$ 0.0094 vs 0.1489 (deviation 2.46 σ)
Duration fit vs published	0.1025 d vs 0.1304 d (deviation 5.45 σ)
Mid-time O–C	5.3 min
Depth SNR	18.2
Residual scatter	0.92 %

Light curve

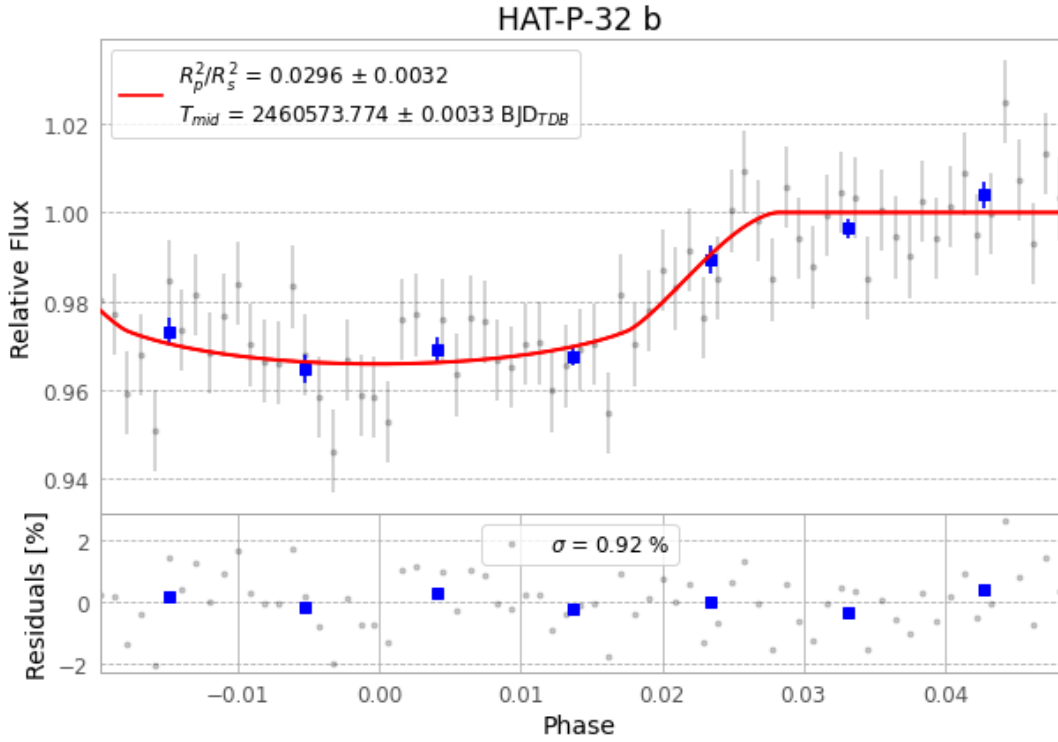


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-09-19.png (SHA-256 be17a070f90a3454...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 195], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6034ea50a20fb4f6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

71 FITS frames (47 MB), HATP-32240920052715.FITS → HATP-32240920085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-240920.log (15fcf2760045...), FinalLightCurve_HAT-P-32 b_2024-09-19.png (be17a070f90a...), FinalLightCurve_HAT-P-32 b_2024-09-19.csv (cc917ac139f6...)

Compute resources

Wall time 75.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:41Z.